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NUTRIENTS, EUTROPHICATION, AND TOTAL PHOSPHOROUS MODELING

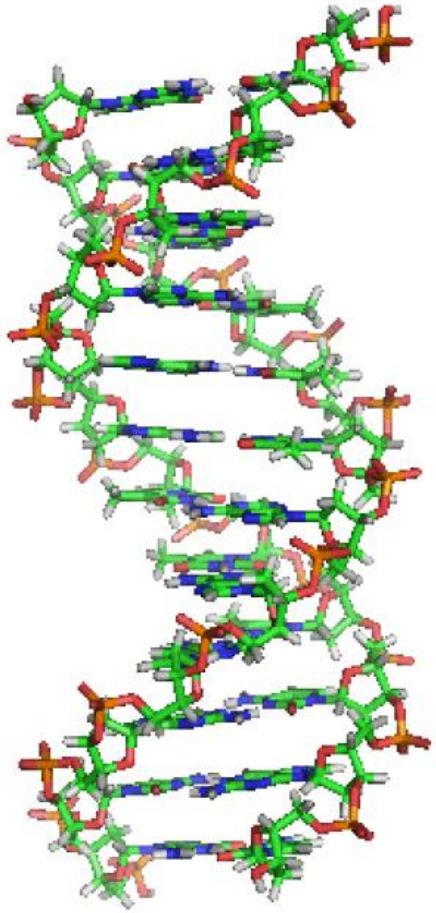
Katrin Schweden

1. Introduction - nutrients

- **Elements**, needed for the growth of plants and organisms

  → e.g., P, N, K, Mg, Ca, …

- They are essential components of **DNA**
- Nutrients are present in the **air, water** and **soil**
- Only **dissolved** nutrients are also plant-available
- Nowadays often efficient to almost complete elimination in wastewater treatment plants (50 – near 100% according to treatment level)



de.wikipedia.org

2. Eutrophication

What:

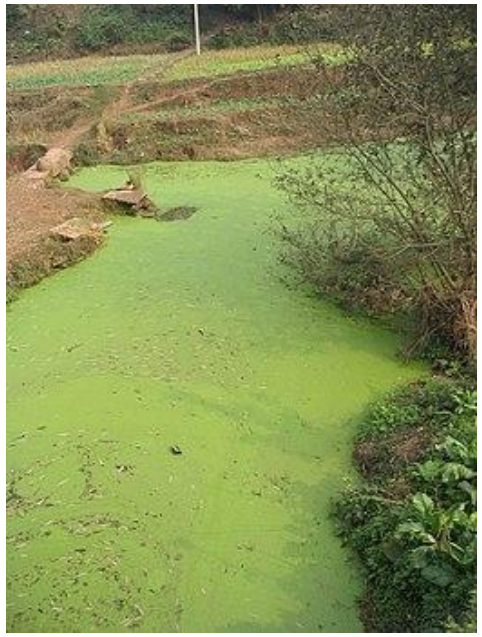
Eutrophication is a **process of excessive nutrient (phosphorus and nitrogen) enrichment** in water bodies leading to excessive plant and algal growth (e.g. algal blooms, Nile cabbage and water hyacinths).

Where:

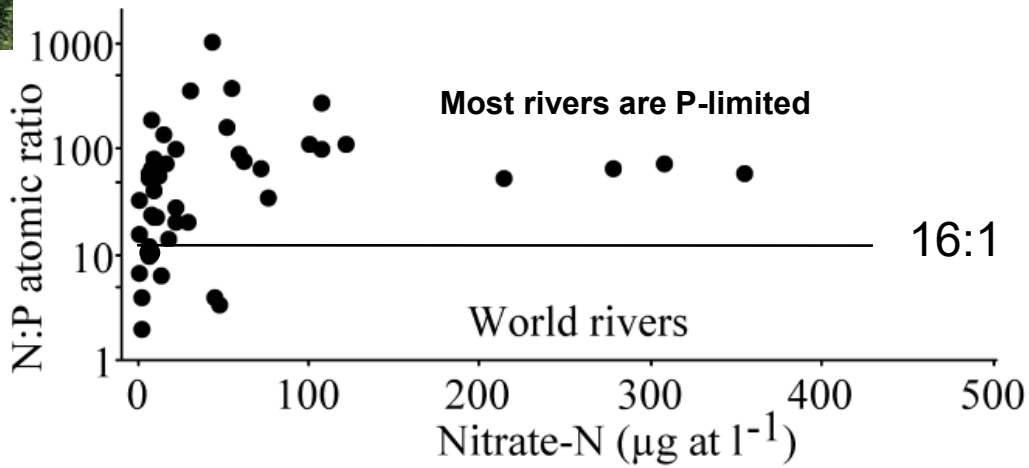
open water sources such as lakes, oceans and reservoirs (Phosphorus is the key nutrient for eutrophication in fresh waters and nitrate is the key substance for salt waters).

Effects:

Excessive algae blocks sun light and depletes oxygen (via own consumption and bacteria decomposition) creating **dead zones** for aquatic life.







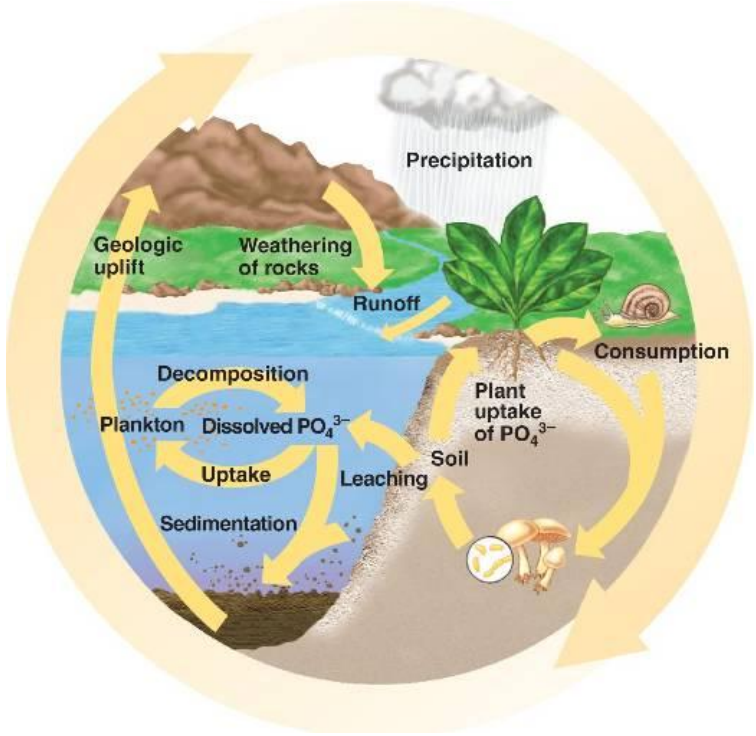
Sources:
www.conserve-energy-future.com
Wikipedia,https://www.eea.europa.eu,
https://earthhow.com/

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2. Eutrophication

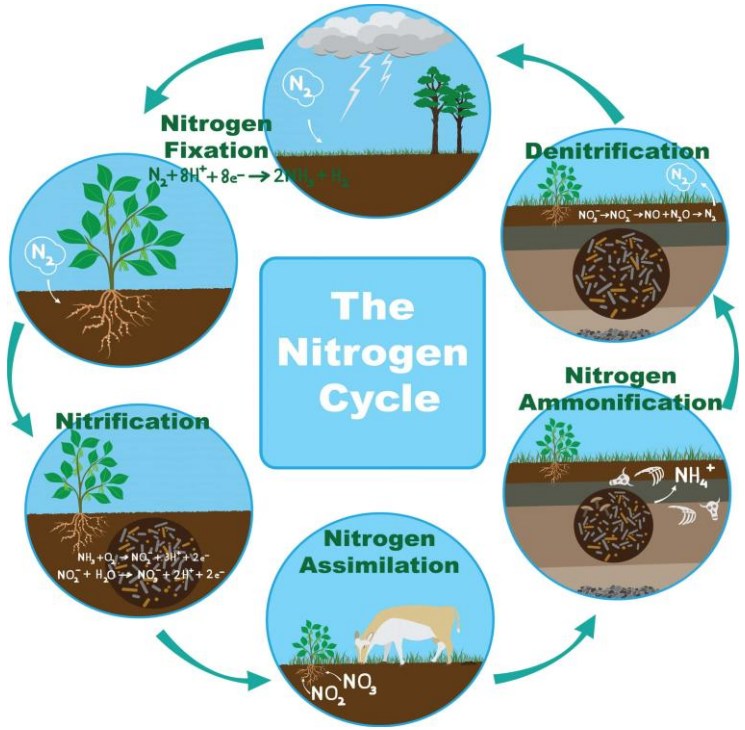
Nutrient cycles:

Phosphorus:



- Phosphorus is an essential nutrient found in the macromolecules of humans and other organisms
- The phosphorus cycle is slow (>>10000 years). Most phosphorus in nature exists in the form of phosphate ions
- Phosphorus is often the limiting nutrient in lakes.

Nitrogen:



- Nitrogen is a key component of the bodies of living organisms
- Nitrogen exists in the atmosphere as N_2 gas. N turnover depends on the specific hyd. cycle
- In nitrogen fixation, bacteria convert N_2 into ammonia, a form of nitrogen usable by plants
- When animals eat the plants, they acquire usable nitrogen compounds
- Nitrogen is a common limiting nutrient in nature, and agriculture

Sources:
Socratic.org,algaebarn.com
khanacademy.org

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2. Eutrophication

Sources of eutrophication:

Source Type	Anthropogenic	Natural
Point	- Sewered households and industrial discharges - Urban surface runoff (if collected) - Concentrated animal feeding operations	
Diffusive	- Fertilizer application - Non-sewered household wastes	- Geogenic background - Natural Events

Point sources:



Diffuse sources:

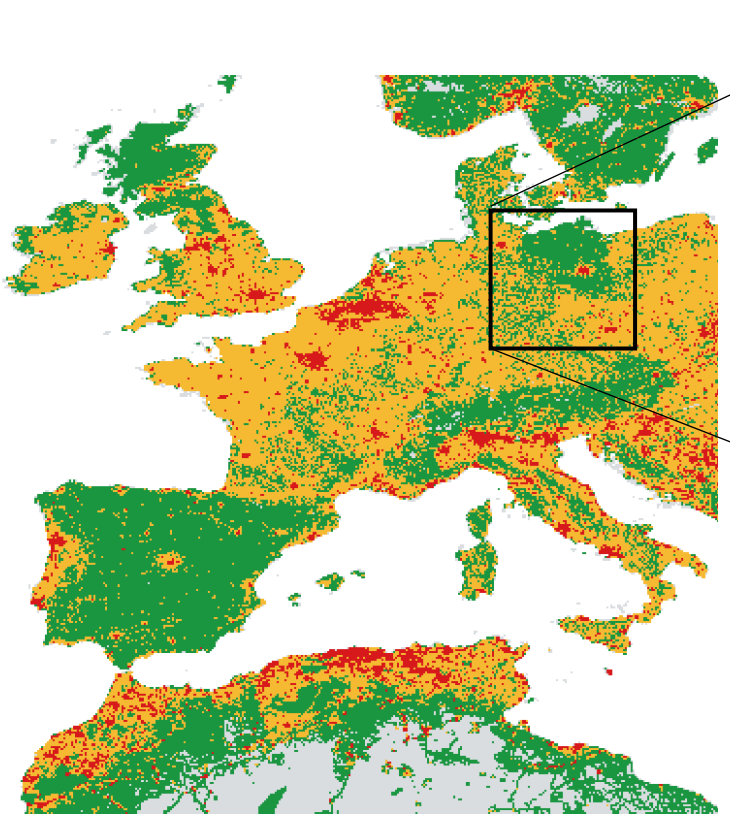


Geogenic background:

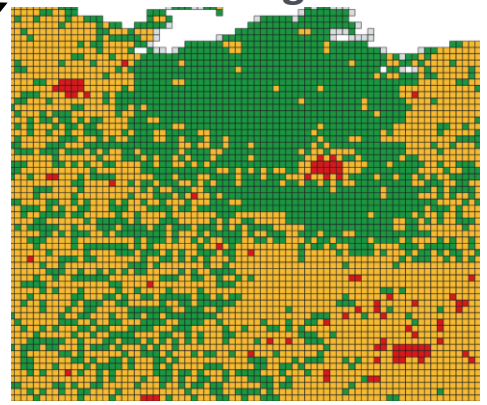
- Atmospheric deposition
- Chemical weathering

Sources:
World Atlas

3. WaterGAP3 Modeling Framework - WorldQual



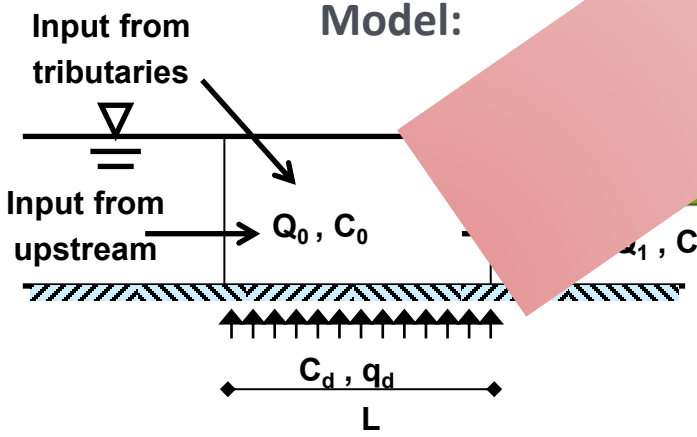
Loadings:



Hydro morphology:

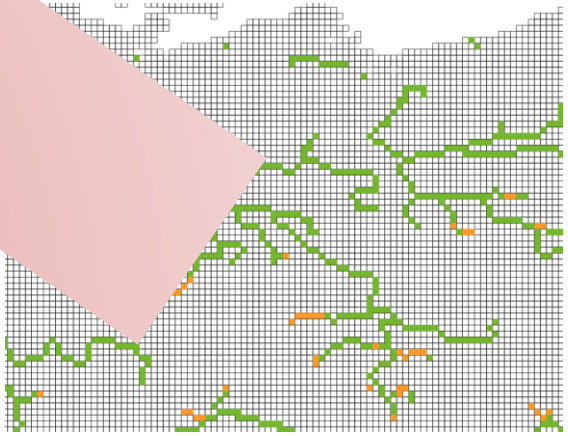


Model:



Input from diffuse and point sources

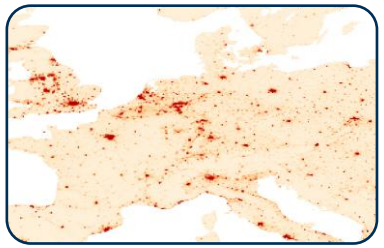
Concentrations:



Voß et al. 2012

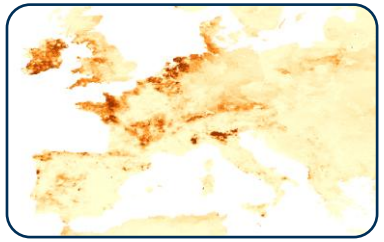
4. WorldQual – Total Phosphorus (TP) loadings

Total Phosphorus loadings model:



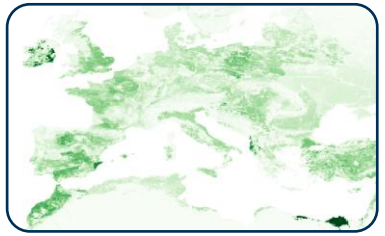
Domestic sector:

- Gridded Population
- Sanitation and treatment
- Emission factor per capita



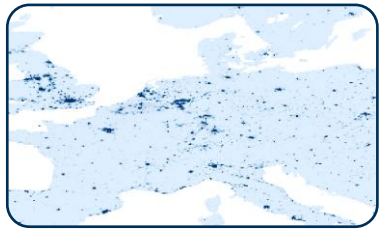
Organic fertilizers:

- Gridded livestock
- Excretion rates per livestock
- Wash off effect (leaching/erosion)



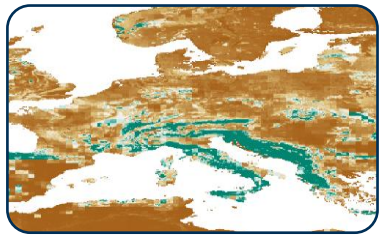
Industrial fertilizers:

- Gridded cropland
- Gridded fertilizer use
- Wash off effect (leaching/erosion)



Manufacturing:

- Gridded return flows
- Effluent concentrations



Background:

- Gridded atmospheric deposition
- Gridded chemical weathering
- Wash off effect (leaching/erosion)



Urban runoff:

- Event mean concentration
- Sanitation and treatment
- Surface runoff

Total Phosphorus Emissions + retention in the water system = export loadings

4. WorldQual – Total Phosphorus (TP) loadings

Total Phosphorus loadings model:

Untreated loadings domestic sewered sector (country):

$$TP_{untrSewered} = EF \times \left(PopUrb \times \frac{ConUrb}{100} + PopRur \times \frac{ConRur}{100} \right)$$

Treated loadings domestic sewered sector (country):

$$TP_{AfterPrimTreatSewered} = \left(\frac{conPrim}{conTotal} \right) \times \left(100 - remPrim \times \frac{stp_{efficiency}}{100} \right) \times \frac{TP_{untrSewered}}{100}$$

$$TP_{AfterSecTreatSewered} = \left(\frac{conSec}{conTotal} \right) \times \left(100 - remSec \times \frac{stp_{efficiency}}{100} \right) \times \frac{TP_{untrSewered}}{100}$$

$$TP_{AfterTertTreatSewered} = \left(\frac{conTert}{conTotal} \right) \times \left(100 - remTert \times \frac{stp_{efficiency}}{100} \right) \times \frac{TP_{untrSewered}}{100}$$

$$TP_{AfterNoTreatSewered} = \left(\frac{conUntr}{conTotal} \right) \times \left(100 - remUntr \times \frac{stp_{efficiency}}{100} \right) \times \frac{TP_{untrSewered}}{100}$$

Sources:
Fink et al. 2018

4. WorldQual – Total Phosphorus (TP) loadings

Total Phosphorus loadings model:

Untreated loadings domestic non-sewered sector (country):

$$TP_{untrNonSewered} = EF \times (PopUrb \times onsite_{conUrb} + PopRur \times onsite_{conRur})$$

Treated loadings domestic non-sewered sector (country):

$$TP_{AfterTreatNonSewered} = onsiteTreat \times TP_{untrNonSewered}$$

Untreated loadings manufacturing sector (country):

$$TP_{untrSewered} = rtf_{manCountry} \times conc_{man}$$

Treated loadings manufacturing sector (country):

$$TP_{AfterTreatSewered} = \frac{TP_{untrSewered}}{10000}\left[\left(100 - remSec \times \frac{stp_{efficiency}}{100}\right) \times frac_{manSec} + \left(100 - remTert \times \frac{stp_{efficiency}}{100}\right) \times frac_{manTert}\right]$$

Sources:
Fink et al. 2018

4. WorldQual – Total Phosphorus (TP) loadings

Total Phosphorus loadings model:

Treated loadings domestic sewered sector (cell):

$$TP_{AfterTreatSewered,cell} = \frac{TP_{AfterTreatSewered,country}}{PopTotCountry \times ConTotCountry} \times \left(PopUrbCell \times \frac{ConUrbCountry}{100} + PopRurCell \times \frac{ConRurCountry}{100} \right)$$

Treated loadings domestic non-sewered sector (cell):

$$TP_{AfterTreatNonSewered,cell} = \frac{TP_{AfterTreatNonSewered,country}}{PopTotCountry \times \left(\frac{100 - ConTotCountry}{100}\right)} \times \left(PopUrbCell \times \frac{100 - ConUrbCountry}{100} + PopRurCell \times \frac{100 - ConRurCountry}{100} \right)$$

Sources:
Fink et al. 2018

4. WorldQual – Total Phosphorus (TP) loadings

Total Phosphorus loadings model:

Treated loadings manufacturing sector (cell):

$$TP_{AfterTreatSewered,cell} = \frac{TP_{AfterTreatSewered,country}}{rtf_{manCountry}} \times rtf_{man,cell}$$

Untreated loadings urban runoff (cell):

$$TP_{untreatUrban,cell} = (UrbanRunoffYear \times cellArea \times builtup) \times conc_{urb}$$

Treated loadings urban runoff (cell):

$$TP_{AfterTreatUrban,cell} = TP_{AfterPrimTreatSewered,cell} + TP_{AfterSecTreatSeweredswxwn,cell} + TP_{AfterTertTreatSewered,cell} + TP_{AfterNoTreatSewered,cell}$$

Sources:
Fink et al. 2018

4. WorldQual – Total Phosphorus (TP) loadings

Total Phosphorus loadings model:

Organic fertilizers/manure (cell):

$$TP_{manure,cell} = TP_{manureProd} \times \left(\frac{0.04}{1 + \left(\frac{surfaceRunoffYear}{800} \right)^{-2}} + SoilEros \times c_{param} \times \frac{surfaceRunoffYear}{surfaceRunoffMean} \right)$$

Inorganic fertilizers/manure (cell):

$$TP_{industrialFertilizer,cell}$$
$$= p_{rate,cell} \times Area_{p-rate,cell} \times \left(\frac{0.04}{1 + \left(\frac{surfaceRunoffYear}{800} \right)^{-2}} + SoilEros \times c_{param} \times \frac{surfaceRunoffYear}{surfaceRunoffMean} \right)$$

$$Area_{p-rate,cell} = cropland \times (1 - builtup)$$

Sources:
Fink et al. 2018

4. WorldQual – Total Phosphorus (TP) loadings

Total Phosphorus loadings model:

Atmospheric deposition (cell):

$$TP_{atmospheric,cell} = TP_{atmosphericInput,cell} \times LandArea_{eff,cell} \times \left(\frac{0.04}{1 + \left(\frac{surfaceRunoffYear}{800} \right)^{-2}} + SoilEros \times c_{param} \times \frac{surfaceRunoffYear}{surfaceRunoffMean} \right)$$

Chemical weathering (cell):

$$TP_{weathering,cell} = TP_{weatheringInput,cell} \times LandArea_{eff,cell} \times \frac{surfaceRunoffYear}{surfaceRunoffMean}$$

Sources:
Fink et al. 2018

4. WorldQual – Total Phosphorus (TP) loadings

Total Phosphorus loadings model:

Total export loadings (basin):

$$TP_{total,year} = retention_{year} \times \sum_{source} TP_{source,year}$$

$$retention = \frac{factor}{1 + factor} \qquad factor = 13.3 \times hydraulicLoad^{-0.93}$$

$$hydraulicLoad = \frac{volSurfaceRunoffYear}{\sum_{cell} wetAreas}$$

Sources:
Hejzlar et al., 2009
Fink et al.2018